Talking Points & Q&A Prep — Hardware vs. Generative Model Efficiency

Oct 27, 2026 · @Zajason

Framing

Open with one sentence and come back to it at the end: **"Papers say a sampler needs fewer steps, but steps are not seconds or joules — I test whether diffusion-vs-flow efficiency conclusions survive a change of GPU."**

What to make obvious in the first two minutes, because that is what you are graded on:

- **Feasibility:** no training, public checkpoints, free or already-available hardware, a staged plan with explicit cuts. Say "no training" out loud early.
- **Connection to the course:** the project is built on core generative-model material — DDPM, score/ODE view of diffusion, DDIM and DPM-Solver as ODE solvers, flow matching, FID and precision/recall. Name the specific lectures these appeared in when you present (fill in the lecture numbers yourself).
- **A falsifiable question:** three hypotheses, each with a named test, and both outcomes are useful.

Target about 8–10 minutes: roughly one minute per slide, less on the title and timeline, more on design and analysis.

Slide-by-slide talking points

1. **Title.** One-line pitch (above). Mention it is inference-only evaluation of pretrained models.
2. **Research question.** Read the boxed question. Then: H1 says NFE predicts latency within one GPU but misses solver overhead, memory bandwidth and precision support. H2 says near-frontier configs can swap rank between devices. H3 says flow matching should lose quality more slowly at low step counts because its paths are straighter. Close with: if rankings turn out stable, that validates NFE as a proxy — also a result.
3. **Why this matters.** Diffusion = iterative denoising; DDIM and DPM-Solver treat sampling as solving an ODE in fewer steps. Flow matching learns the velocity field directly. DiT and SiT are the controlled pair: same transformer backbone, different

training objective, so differences come from the objective and sampler, not the network. The gap: efficiency is reported as NFE or one-GPU timings, and FID itself is sensitive to evaluation details, so the protocol must be fixed.

4. **Experimental design.** 24 = 2 models × 2 samplers each × 3 step counts × 2 precisions. Batch size 1 vs 8 only for timing (latency-bound vs throughput-bound regime). Fixed: guidance scale, compilation, seeds, class labels. Every network evaluation counts toward NFE — both CFG passes and Heun's second evaluation per step.
5. **Hardware and protocol.** Three devices across three architecture generations, very different bandwidth and precision support. One GPU per run so GH200 numbers are comparable. Pinned versions, warm-up, CUDA-event timing, NVML memory and energy, medians with IQR because GPU timings have long tails.
6. **Quality evaluation.** Quality is a property of model + sampler + precision, not of the GPU, up to numerical noise — so generate images once and verify that assumption on selected outputs per device. Pilot at 2k images, promote only near-frontier configs to 10k × 3 seeds.
7. **Analysis.** One test per hypothesis. Stress matched measured cost for H3: matching on steps or NFE would be unfair to Heun and to multistep solvers; milliseconds and joules are what a practitioner pays.
8. **Feasibility.** Read the left column fast, then the fallback order: drop fp32 quality generation first, then Ikaros to batch 1. The research question survives every cut; only breadth is lost.
9. **Schedule.** Name three checkpoints: harness validated by week 4, pilot frontier by week 7, cross-device data by week 9.
10. **Contribution.** Three deliverables, then the "either outcome is informative" line, then invite questions.

Likely questions and how to answer

Short answer first, then one supporting detail. If you don't know, say what you would measure to find out.

Feasibility

"How much compute is this, really?" Have a back-of-envelope ready. A DiT-XL/2 forward pass at 256×256 is roughly 119 GFLOPs. The pilot is 24 configs × 2,000 images = 48k images; with CFG doubling evaluations and Heun adding more, call it ~50 forward passes per image on average → about 3×10^17 FLOPs. That is a few hours to a day on a modern consumer GPU, more for the fp32 half. The 10k × 3-seed stage costs several times that, which is exactly why only near-frontier configs are promoted. (Approximate — run 100 images on your own GPU and quote the real number.)

"What is your local GPU?" The proposal never names it. Know the model, VRAM and whether it supports bf16. XL/2 weights are about 2.7 GB in fp32, so memory is not the bottleneck.

"What if you lose Ikaros access?" The core comparison still runs on two architectures (local + T4). Ikaros adds the modern-datacenter point; fallback is batch 1 only.

"Why mention fine-tuning at all?" Only as insurance if a sampler doesn't plug into a checkpoint cleanly. DiT works with DDIM/DPM-Solver through its noise prediction, and SiT ships with ODE samplers, so you expect zero training.

Connection to the course

"How is this AI and not just benchmarking?" The research question is about generative models: whether the straighter paths of flow matching translate into better quality at low cost, and whether the field's standard efficiency metric (NFE) is valid. Hardware is the experimental variable, not the topic.

"Explain the difference between diffusion and flow matching." Diffusion trains a network to predict noise (or score) under a fixed noising process; sampling solves a reverse SDE/ODE. Flow matching regresses a velocity field along a chosen probability path (here SiT's interpolant) and samples by integrating an ODE. Straighter paths mean less discretisation error per step, hence H3.

"Why DiT and SiT?" Same transformer backbone and size, same dataset and latent space, so objective and sampler are the only changed factors. A controlled comparison.

Methodology

"How do you count NFE?" Every network evaluation: CFG doubles it, Heun uses two evaluations per step (one fewer on the last step in some implementations), DPM-Solver multistep uses one per step. Be ready to state the exact NFE for each of the 24 configs.

"Is the VAE decoder in your latency?" It should be reported both ways: end-to-end latency includes decode, but decode is not an NFE. Its fixed cost is one reason NFE underpredicts latency at low step counts — a nice concrete H1 example.

"T4 has no native bf16 — what do you run there?" Decide before presenting. Clean answer: on T4 use fp16 as the reduced-precision setting and say so, or run bf16 and report the slowdown as an H1 finding (precision support breaking NFE). Don't leave it open.

"Is NVML energy comparable across devices?" Not perfectly. On GH200 the module power can include the Grace CPU and memory, and on Colab the energy counter may be unavailable inside the VM, in which case integrate sampled power over time. Say you will document what each device's counter covers and compare trends within a device, not absolute joules across devices.

"FID with 2,000 images is biased." Yes — biased upward and noisy. It is used only to rank configurations against each other at the same N; final claims use 10k × 3 seeds. You can add KID, which is unbiased at small N, as a pilot cross-check.

"Kendall's τ over a handful of configs has little power." Agree. Report it with a bootstrap interval and also list the specific pairwise frontier swaps, which are the practically meaningful result.

"Won't timing noise swamp small differences?" Warm-up, repeated CUDA-event runs, medians with IQR, fixed clocks where the platform allows. Colab is shared, so treat T4 IQRs as wider and only claim swaps outside the IQR.

"Why ImageNet-256 and not text-to-image?" Public matched checkpoints, standard FID reference statistics, and affordable. Text-to-image adds a text encoder and prompt-set choices that would blur the comparison.

Fix or own before presenting

- [ ] Name your local GPU (model, VRAM, bf16 support) — it is the workhorse and the proposal never says what it is.
- [ ] Decide the T4 reduced-precision setting (fp16 substitute, or bf16 with the slowdown reported).
- [ ] Time 100 images on your own GPU so the compute estimate is a measured number, not a guess.
- [ ] Write down the exact NFE per configuration, including CFG and Heun's extra evaluations.
- [ ] Check whether NVML energy works on Colab and what the GH200 counter covers.
- [ ] Map each slide-3 concept to the lecture where the course covered it.
- [ ] In the written proposal, the citations render as [?] (broken bibliography keys) and the schedule heading reads "PROPOSEDSCHEDULE" — fix before resubmitting.